

Activity Configuring and Securing a Wireless LAN in Packet Tracer

Objectives:

1. Configure a Wireless Access Point (WAP) local IP address
 2. Configure a WAP with an SSID
 3. Change the administrator's password on a WAP
 4. Configure a WAP to use the WPA2 protocol
 5. Configure the DHCP server on a WAP
 6. Configure a MAC filter on a WAP
 7. Configure a DHCP Reservation
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1. Open the file "Activity Configuring and Securing a Wireless LAN in Packet Tracer.pka"
 2. Open PC_A and select the Desktop tab.
 3. Select the command prompt and check the current IP address with the **ipconfig /all** command.
Record the IP address here: ____0____.____0____.____0____.____0____
Subnet mask: ____0____.____0____.____0____.____0____
 4. What is the MAC address of PC_A? 000A:4140:595C
 5. Close the command prompt and select the IP Configuration window.
 6. Set the address to DHCP.
 7. Select the command prompt and check the current IP address with the **ipconfig /all** command. Record the address here: 192.168.0.105 mask: 255.255.255.0
 8. What is the default gateway address? 192.168.0.1
 9. What device in the network is the default gateway?
_____router_____
 10. What is the IP address of the DHCP server? 192.168.0.1
 11. Open the web browser and access the WAP by opening <http://192.168.0.1>
 12. The default login information for the WAP is:
User Name: admin
Password: admin
 13. You should now be at the configuration window for the WAP.
 14. What is the Internet Setup IP address assignment?
_____DHCP_____
 15. What is this connection used for? ____It's used to automatically obtain an IP address from the ISP_____
 16. What is the default Network Setup IP address? 192.168.0.1
 17. Determine the fifth private class B network and borrow 8 bits for subnets.

Name: _____

list the first 3 subnets created:

1. 172.20.0.0 /24
2. 172.20.1.0 /24
3. 172.20.2.0 /24

18. How many subnets have been created? _____ 256 subnets _____
19. How many hosts are on each of the subnets? _ 254 hosts per subnet _____
20. Assign the local network the first IP address on the second newly created subnet. What is the new IP address of the router? _____ 172 _____ . _____ 20 _____ . _____ 1 _____ . _____ 1 _____
21. Start the DHCP server on the 15-th IP address and set the maximum number of users to be 25.
22. What is the range of IP address that will be used for DHCP clients?
_____ 172 _____ . _____ 20 _____ . _____ 1 _____ . _____ 15 _____ through
_____ 172 _____ . _____ 20 _____ . _____ 1 _____ . _____ 39 _____
23. Scroll to the bottom and select Save Settings.
24. What happened to the connection? _____ request timeout _____
25. What caused this to happen? _____ The connection dropped because I changed the router's IP address and network. _____
26. Close the web browser.
27. Open the command prompt on PC1 and request a new IP address assignment from the DHCP server with the **ipconfig /renew** command.
28. What address did PC_A obtain this time? _____ 172 _____ . _____ 20 _____ . _____ 1 _____ . _____ 105 _____
29. Close the command prompt window.
30. Open the web browser on PC_A and access the WAP.
31. Select the Wireless menu.
32. Configure the following wireless networks:

2.4 GHz

Network Mode:	Auto
Network Name (SSID):	WLAN24
SSID Broadcast:	☐ Enabled ☒ Disabled
Standard Channel:	1 - 2.412GHz
Channel Bandwidth:	Auto

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Network Mode:	Auto
Network Name (SSID):	WLAN52
SSID Broadcast:	☒ Enabled ☐ Disabled
Standard Channel:	Auto
Channel Bandwidth:	Auto

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Network Mode:	Auto
Network Name (SSID):	WLAN51
SSID Broadcast:	☒ Enabled ☐ Disabled
Standard Channel:	Auto
Channel Bandwidth:	Auto

33. What does SSID stand for? _____ Service Set Identifier _____
34. If a WAP is not broadcasting its SSID, can clients still connect, and how? __ Yes, clients can still connect by manually entering the SSID and the network's security credentials. _____
35. Scroll to the bottom and select Save Settings.
36. Access Laptop_1 and select the Desktop tab.
37. Select PC Wireless and the connect tab. Do you see the WLAN24 network? __ No _____
38. Use the Config tab in PT and connect to the WLAN24 network.
39. Access Laptop_2 and connect to the WLAN51 network. **If you have any trouble connecting or seeing a WLAN, you may need to update the wireless module. Do that by shutting down the laptop and removing the Ethernet adapter. Add the following module: PT-LAPTOP-NM-1W-AC module.
40. Access Laptop_3 and connect to the WLAN51 network.
41. Access Laptop_4 and connect to the WLAN52 network.

Name: _____

42. Next create a wireless MAC address filter. Access Laptop_1, open the desktop and use the command prompt to determine the MAC addresses and record it here: 00D0:FF50:E5B1
43. What is the Block ID of Laptop1? ____00____:____D0____:____FF____
44. What is the device ID of Laptop1? ____50____:____E5____:____B1____
45. Move back to the GUI for the WAP.
46. Select Wireless > Wireless MAC Filter.
47. Enable the Wireless MAC filter for the 2.4G Wireless Port and select the option to Permit PC's listed below to access the wireless network.
48. Enter the MAC address of Laptop_1, be sure to follow the proper format for the MAC addresses, 2 digits followed by a colon.
49. Scroll to the bottom and select Save Settings.
50. Verify that Laptop_1 can connect to the WLAN24 network.
51. Next create a DHCP reservation. Access Laptop_3, open the desktop and use the command prompt to determine the MAC addresses and record it here:
- ____00____:____E0____:____8F50____:____5B____:____47____
52. What is the Block ID of Laptop_3? ____00____:____E0____:____8F____ 53. What is the device ID of Laptop_3? ____50____:____5B____:____47____
54. Move back to the GUI for the WAP.
55. Select Setup > Basic Setup > DHCP Reservation.
56. Create a reservation for Laptop_3's MAC address and assign the sixth IP address within the pool, and use the name Admin Laptop.
57. Open Laptop_3 and renew its IP address. What IP address did it obtain?
- ____172____.____20____.____1____.____21____
58. The last step in configuring this wireless network is to configure security.
59. Move back to the GUI for the WAP and select Wireless > Wireless Security.
60. What options are available for the Security Mode?
- ____WEP____
- ____WPA Personal____
- ____WPA Enterprise____
- ____WPA2 Personal____
- ____WPA2 Enterprise____
61. Which of the options is the LEAST preferred security mode? ____WEP____
62. Which of the options is the NEXT LEAST preferred security mode? ____WPA Personal____
63. Set security to WPA2 Personal.
64. What are the two options under Encryption?

Name: _____ Daphny Forcho _____

_____ AES _____ TKIP _____

65. Set encryption to AES.
66. What does AES stand for? _____ Advance Encryption
Standard _____
67. Configure the networks as follows:

2.4 GHz

Security Mode:	WPA2 Personal	▼
Encryption:	AES	▼
Passphrase:	P@ssw0rd24	
Key Renewal:	3600	seconds

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Security Mode:	WPA2 Personal	▼
Encryption:	AES	▼
Passphrase:	P@ssw0rd51	
Key Renewal:	3600	seconds

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Security Mode:	WPA2 Personal	▼
Encryption:	AES	▼
Passphrase:	P@ssw0rd52	
Key Renewal:	3600	seconds

68. Scroll to the bottom and select Save Settings.
69. Move back to the Packet Tracer window.
70. What happened to your connection between the Laptops and the WAP? _____ they all
disconnected _____
71. Open Laptop_1 and select the Config tab.
72. Select the wireless button and under the Authentication section select WPA2-PSK. What does PSK stand
for? _____ Pre Shared Key _____
73. Enter the PassPhrase that you configured.
74. Ensure that the Encryption Type is set to AES.
75. Move back to the Packet Tracer window.
76. What happened to your connection between the Laptop and the WAP? _____ It connected

Name: _____

77. Move back to the GUI for the WAP and select the Administration section.
78. Set the router password to cisco.
79. Scroll to the bottom and select Save Settings.
80. Move back to the packet Tracer main window.
81. Reconnect Laptop2, Laptop3 and Laptop4 to the wireless network.
82. Set PC_B to obtain an IP address through DHCP.
83. Configure the following Guest network:

☐ Allow guests to see each other and access the local network

2.4 GHz

☐ Enable Guest Profile

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☒ Enable Guest Profile

Network Name (SSID): Guest50

☒ Broadcast SSID

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: Guest@50

Key Renewal: 3600 seconds

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☐ Enable Guest Profile

84. Connect the smart phone to the newly created guest network.
85. Ensure all end devices are connected to a wireless network.